

COLONIA DEL SACRAMENTO, Uruguay

WMO: 865600

Lat: 34.452S

Lon: 57.767W

Elev: 20

StdP: 101.08

Time Zone: -3.00 (W03)

Period: 94-02

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.9	5.3	-2.0	3.2	6.8	-0.5	3.6	7.8	15.2	9.6	13.9	9.3	3.3	60	0.481

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.7	31.2	23.0	30.0	22.5	28.8	22.1	24.8	29.2	24.0	28.2	23.2	27.0	4.3	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.4	18.2	27.6	22.7	17.4	26.8	21.9	16.6	25.8	75.7	29.3	72.4	28.3	69.3	27.0	28.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.7	11.6	10.1	1.3	35.3	0.7	1.7	0.8	36.5	0.4	37.5	0.0	38.4	-0.5	39.6		
(5)			0.3	27.0	1.0	0.9	-0.4	27.7	-1.0	28.3	-1.6	28.8	-2.4	29.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.6	23.3	22.8	21.6	18.0	15.3	12.1	11.2	13.3	14.3	17.3	19.9	22.4
	DBStd	5.14	2.46	2.42	2.55	2.66	3.02	3.33	3.32	3.75	3.01	2.86	2.98	2.87
	HDD10.0	47	0	0	0	0	1	14	22	8	2	1	0	0
	HDD18.3	926	1	1	4	37	104	191	224	166	126	53	16	3
	CDD10.0	2823	413	357	361	240	164	76	60	109	132	228	297	386
	CDD18.3	659	156	125	106	27	9	3	3	9	6	22	63	131
	CDH23.3	4021	1164	774	506	73	14	3	9	59	35	94	357	933
	CDH26.7	936	299	175	92	7	1	0	1	15	6	17	79	245
Wind	WSAvg	4.8	4.6	4.8	4.8	4.9	4.2	4.8	4.7	4.9	4.9	4.9	4.9	5.1
Precipitation	PrecAvg	1147	111	128	119	102	77	69	71	70	80	111	115	107
	PrecMax	1746	238	342	486	247	304	231	157	205	196	310	328	355
	PrecMin	642	29	30	25	13	9	9	10	10	14	11	28	3
	PrecStd	271	57	87	92	65	59	46	39	54	52	71	65	83
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.7	32.2	31.2	27.6	25.1	23.4	24.7	28.7	27.1	29.1	31.2	33.0
		MCWB	24.4	24.0	23.1	21.9	19.7	18.5	18.8	21.4	20.7	21.2	21.9	23.7
	2%	DB	30.8	30.1	29.1	25.2	22.8	20.8	20.5	24.3	23.7	25.7	28.9	30.7
		MCWB	23.3	22.9	22.7	19.9	18.5	17.8	16.7	18.7	17.9	19.4	20.4	22.4
	5%	DB	29.6	28.7	27.3	23.8	21.1	18.7	17.6	21.5	21.1	23.9	27.0	29.1
		MCWB	22.6	22.4	21.8	19.2	17.8	15.9	14.7	17.1	15.8	18.2	20.3	22.1
	10%	DB	28.3	27.4	26.0	22.2	19.9	16.8	15.7	18.8	19.0	22.0	25.3	27.7
		MCWB	22.0	21.8	21.2	18.8	17.0	14.8	13.1	14.7	14.7	17.5	19.2	21.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.1	25.2	25.4	23.1	21.0	19.4	18.8	21.5	21.8	22.7	23.8	25.6
		MCDB	30.6	29.8	29.8	26.0	24.1	21.6	24.1	27.8	25.4	26.7	28.8	30.7
	2%	WB	24.8	24.2	24.1	21.3	19.4	17.9	17.0	19.1	18.8	20.8	22.3	24.2
		MCDB	29.0	28.6	27.0	23.9	21.7	20.0	20.0	23.5	23.2	23.8	26.4	29.1
	5%	WB	23.8	23.3	23.0	20.3	18.3	16.6	15.2	17.4	16.7	19.6	21.1	23.1
		MCDB	28.1	27.4	25.6	22.5	20.4	18.5	16.7	21.0	19.3	22.0	25.2	27.3
	10%	WB	23.0	22.5	22.1	19.4	17.4	15.1	13.7	15.7	15.6	18.8	20.1	22.2
		MCDB	26.9	26.4	24.9	21.5	19.4	16.3	15.2	18.1	17.9	21.1	24.4	26.4
Mean Daily Temperature Range	5% DB	MDBR	8.7	8.0	7.4	6.7	6.3	5.8	6.3	7.3	7.5	7.6	8.8	8.7
		MCDBR	10.4	9.9	9.4	9.0	7.8	8.1	8.9	10.9	11.0	10.7	11.3	11.2
		MCWBR	6.1	5.8	5.3	5.8	5.4	5.7	5.5	6.4	6.7	6.2	6.1	6.9
	5% WB	MCDBR	8.8	8.3	7.6	6.9	6.9	6.8	7.3	10.2	8.6	8.0	9.4	9.1
		MCWBR	6.1	5.7	4.9	5.2	5.1	5.4	5.2	6.5	6.0	5.9	6.2	6.2
Clear-Sky Solar Irradiance	taub	0.406	0.396	0.380	0.381	0.368	0.365	0.359	0.439	0.461	0.423	0.394	0.398	
	taud	2.400	2.446	2.475	2.412	2.422	2.423	2.412	2.134	2.090	2.268	2.386	2.398	
	Ebn at Noon	932	919	897	832	785	757	785	754	800	882	937	943	
	Edh at Noon	126	116	105	100	87	82	87	129	152	137	127	127	
All-Sky Solar Radiation	RadAvg	7.30	6.34	5.31	3.85	2.68	2.27	2.36	3.19	4.39	5.62	6.81	7.55	
	RadStd	0.43	0.59	0.49	0.43	0.22	0.22	0.26	0.32	0.30	0.52	0.50	0.39	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)		+0.35	N/A	N/A	+0.64	+0.44	N/A	N/A	+106	+57

Nomenclature: See separate page